CIS6003 Advanced Programming — Assignment Report

Assessment: Online Vehicle Reservation System (WRIT1) | **Weighting:** 100%
Student: BSc SE – CIS-6003 – 20374265 **Submission title format:** `st20374265 CIS6003 WRIT1` **GitHub repository:** https://github.com/TharusanK23/vehicle-reservation-system

> **Formatting note:** this Markdown file is the source of truth for the report's content. A submission-ready PDF (`docs/ASSIGNMENT_REPORT.pdf`), already formatted to the brief's spec (A4, margins 1.5in/1in, 1.5 line spacing, Times New Roman, 14pt bold headings, 12pt body, page numbers bottom-right) with every diagram and screenshot embedded below, is generated directly from this file — see `docs/generate-pdf.js`.

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1. Introduction

Sunrise Vehicle Rentals is a fictional vehicle rental branch that — mirroring the operational problems described in the assignment brief's scenario — currently manages bookings on paper, resulting in double bookings, lost customer records, and billing errors. This report documents the design, development, testing, and evaluation of a computerised, web-based **Online Vehicle Reservation System** built to solve exactly those problems, developed end-to-end using Java (Spring Boot) for the back end, a static HTML/CSS/ JavaScript client for the front end, and MySQL (via XAMPP) for persistence.

1.1 A note on the brief's scenario text versus its title

The assignment brief's **title** and the coursework's **file name** both say "Online Vehicle Reservation System", but the brief's **scenario narrative** describes a dental clinic's appointment book (patients, dentists, treatment types, consultation fees). This is evidently a template-reuse artefact in the brief document rather than an intentional requirement. The brief explicitly states: *"Students are free to make necessary assumptions on system design ... but all suggestions must be well explained with valid reasons"* (Assessment Brief, p.3). Accordingly, this project implements the vehicle-reservation domain that the title and coursework naming specify, while preserving **every functional requirement from the scenario text**

one-for-one by mapping each dental-clinic concept onto its vehicle-rental equivalent:

Brief wording	This system
Unique appointment number	Unique reservation number (`RES-<year>-<sequence>`)
Patient (name, address, contact number)	Customer (name, address, contact number, email, licence number)
Dentist name	Vehicle (registration number, make, model, year, category)
Treatment type	Vehicle category / rental package (Economy, Sedan, SUV, Van, Luxury)
Appointment date and time	Pickup date and time
(not present)	Return date and time — added because a *rental* inherently needs a return leg, whereas a single clinic visit does not; this is exactly the kind of low-risk, brief-permitted extension the "Additional functionalities can be included as needed" clause anticipates
Consultation fee → bill	Category daily rate × rental days → bill

No functionality described in the brief was dropped; every one of the six numbered functionalities (Login, Register, Display, Calculate & Print Bill, Help, Exit) is implemented, evidenced in §5.8 and validated in §6.

1.2 Report structure

This report follows the brief's task lettering: **Task A** (system design with UML, §2), **Task B** (interactive system, design patterns, architecture, database, §3), **Task C** (testing, §4), and **Task D** (Git/GitHub, §5), followed by an explicit self-evaluation against the Excellent-band marking criteria (§6), an EDGE reflection (§7), and a conclusion (§8).

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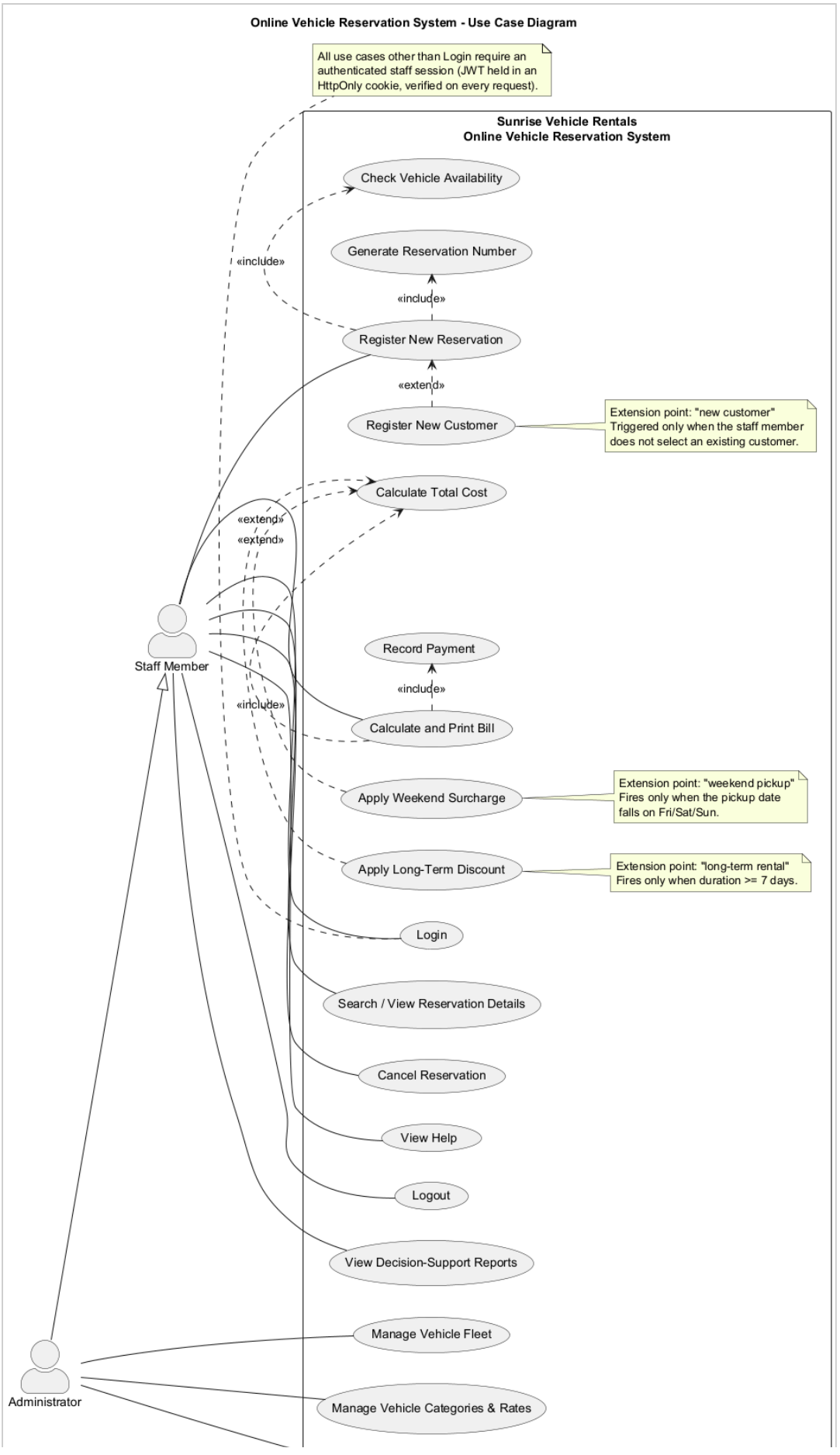
2. Task A — System Design with UML Diagrams (20 marks)

Full-resolution diagrams are in `diagrams/*.png`, generated from version-controlled PlantUML source (`diagrams/*.puml`) so they can be regenerated and audited rather than treated as static images. `diagrams/README.md` documents every

non-obvious design assumption in detail; the key points are summarised here with the reasoning behind them.

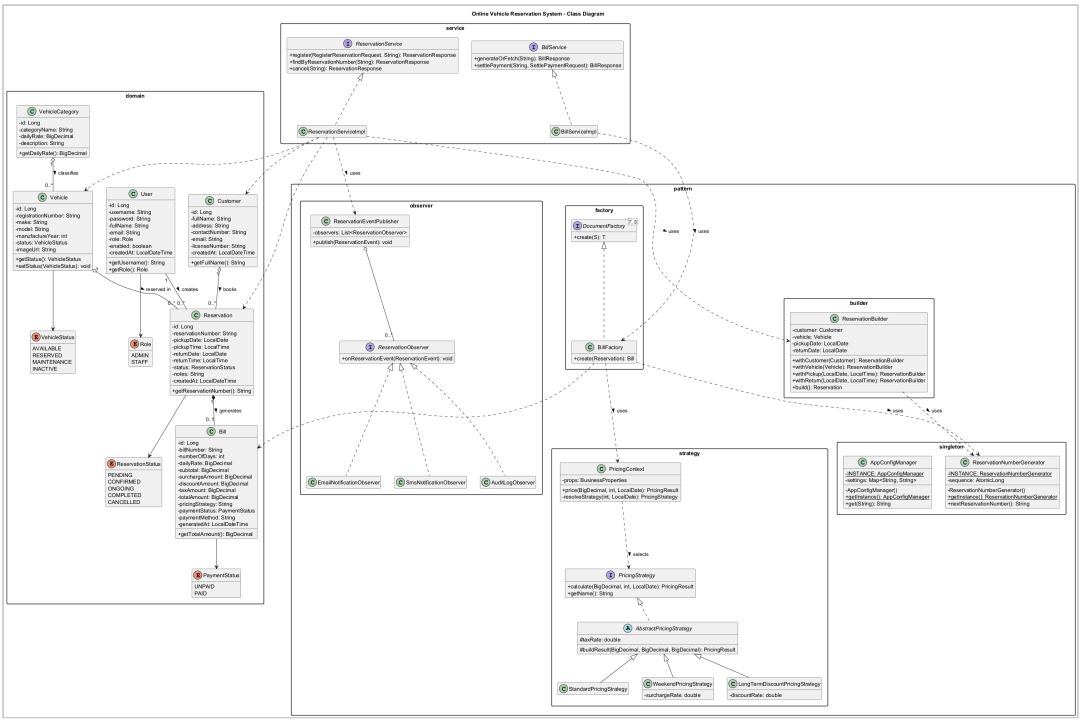
2.1 Use Case Diagram


Online Vehicle Reservation System - Use Case Diagram
All use cases other than Login require an authenticated staff session (JWT held in an HttpOnly cookie, verified on every request).
Sunrise Vehicle Rentals
Online Vehicle Reservation System
Check Vehicle Availability
Generate Reservation Number
«include»
«include»
Register New Reservation
«extend»
Register New Customer
Extension point: "new customer" Triggered only when the staff member does not select an existing customer.
Calculate Total Cost
«extend»
«extend»
Staff Member
Record Payment
«include»
«include»
Calculate and Print Bill
Apply Weekend Surcharge
Extension point: "weekend pickup" Fires only when the pickup date falls on Fri/Sat/Sun.
Apply Long-Term Discount
Extension point: "long-term rental" Fires only when duration >= 7 days.
Login
Search / View Reservation Details
Cancel Reservation
View Help
Logout
View Decision-Support Reports
Manage Vehicle Fleet
Administrator
Manage Vehicle Categories & Rates

Two actors were modelled: **Staff Member** and **Administrator**, related by generalisation (`Administrator --|> Staff`) because every capability available to Staff is also available to an Administrator, who additionally manages the vehicle fleet, categories, and staff accounts. This reflects a considered, brief-permitted assumption: the brief only says *"only authorised staff can use the system"* without specifying roles, but distinguishing operational staff from an administrator is realistic for any multi-user business system and is explicitly rewarded by the Excellent-band criteria ("more sophisticated data representation... separate UI windows").

Two genuine `<<include>>` relationships were modelled — *Register New Reservation* always includes *Check Vehicle Availability* and *Generate Reservation Number*, because these steps unconditionally happen every time a reservation is created — alongside three genuine `<<extend>>` relationships, used correctly for **conditional** behaviour rather than as a synonym for "include": *Register New Customer* extends *Register New Reservation* only when no existing customer is selected; *Apply Weekend Surcharge* and *Apply Long-Term Discount* extend *Calculate Total Cost* only under their respective trigger conditions (pickup date falls on Friday/Saturday/Sunday; rental duration is seven days or more). Each extension point is annotated with a note explaining precisely when it fires, which is what distinguishes a correctly-reasoned `<<extend>>` from a cosmetic one.

2.2 Class Diagram



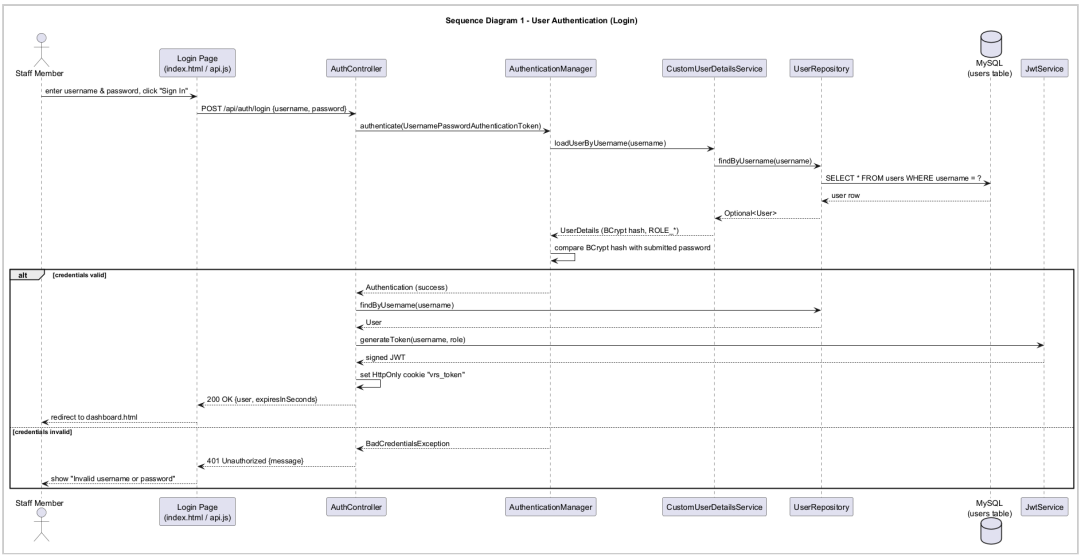
The class diagram is deliberately organised into UML packages that mirror the actual Java package structure (`domain`, `service`, and one package per design pattern — `pattern.builder`, `pattern.factory`, `pattern.strategy`, `pattern.observer`, `pattern.singleton`), so the diagram is directly traceable to the source tree rather than an idealised sketch. Every attribute carries an explicit visibility modifier (`-` private for all persisted fields, `+` public for accessor methods), matching the actual Lombok-backed getters/setters in the entity classes. Relationships carry multiplicity and the correct UML semantics:

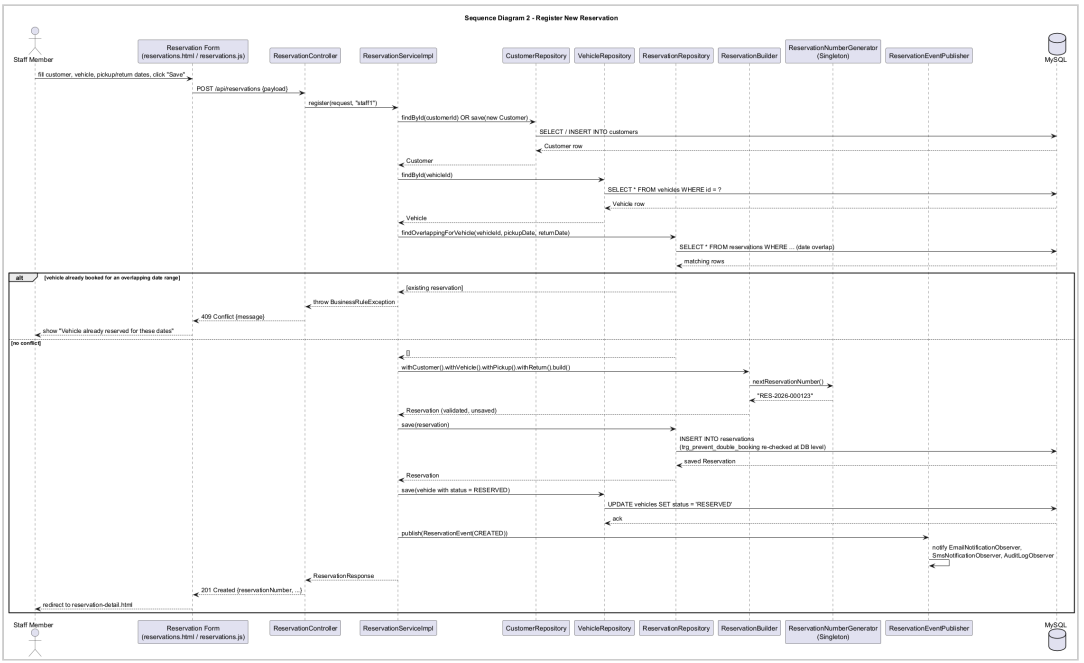
- **Aggregation** (`o--`) is used between `Customer` / `Vehicle` and `Reservation`: a `Reservation` references a `Customer` and a `Vehicle`, but neither is *owned* by the reservation — deleting a reservation must not delete the customer or the vehicle it referenced.
- **Composition** (`*--`) is used between `Reservation` and `Bill` (0..1): a `Bill` cannot meaningfully exist without its `Reservation` and is deleted along with it in this system's lifecycle.
- Interfaces are marked `<<interface>>`/italicised (`ReservationObserver`, `PricingStrategy`, `DocumentFactory`) with realisation arrows to their concrete implementations, and the abstract class `AbstractPricingStrategy` is shown with `AbstractPricingStrategy` in italics per UML convention.

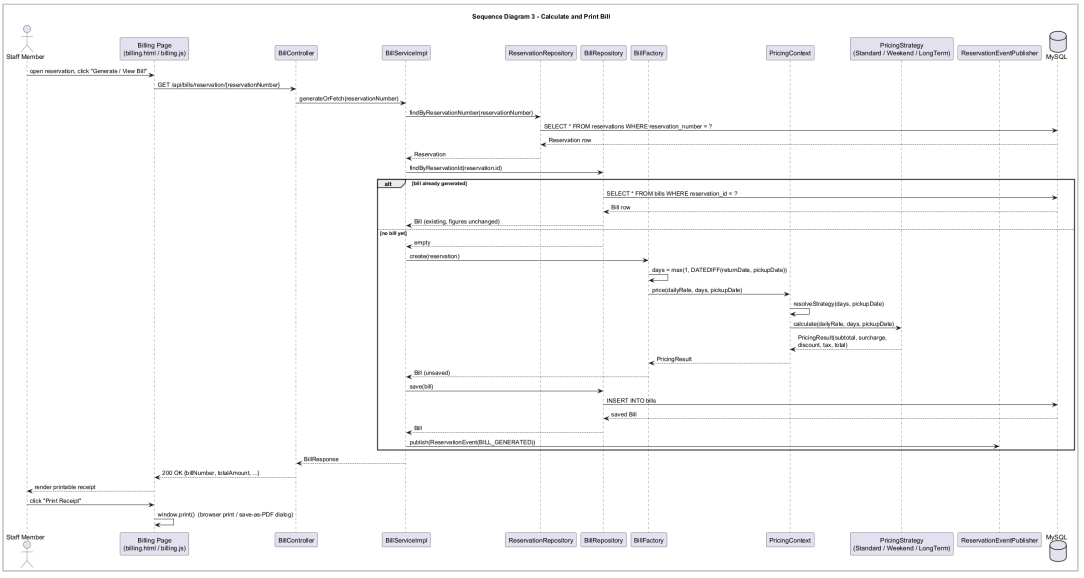
2.3 Sequence Diagrams

Three sequence diagrams were produced, each covering one of the assignment brief's core functionalities end-to-end, from the browser through every architectural layer down to the database — deliberately chosen because these are the three flows a grader is most likely to exercise manually to validate the system, so the diagrams double as an accurate map of what actually happens when they do.

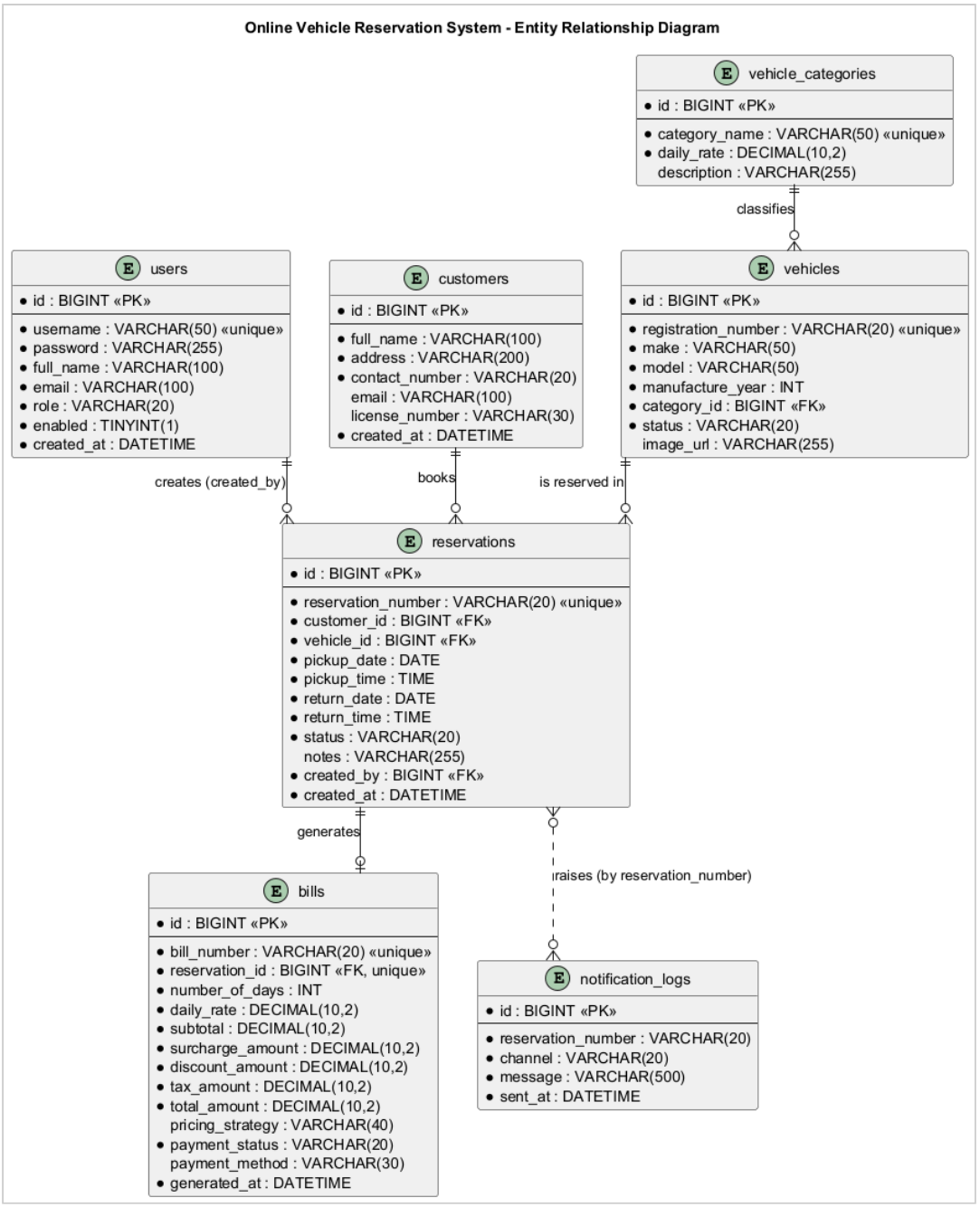
1. **`sequence-01-login.png`** — User Authentication (Login). Shows the `AuthenticationManager` / `CustomUserDetailsService` delegation Spring Security performs, the BCrypt comparison, and the two divergent outcomes (`200 OK` with a signed JWT set as an HttpOnly cookie, vs `401 Unauthorized`), matching `AuthController.login()`.
2. **`sequence-02-register-reservation.png`** — Register New Reservation. Shows customer resolution (existing vs newly-created), the vehicle availability/overlap check, the `ReservationBuilder` assembling a valid `Reservation`, the `ReservationNumberGenerator` singleton minting the reservation number, persistence, the vehicle status transition to `RESERVED`, and the `ReservationEventPublisher` notifying all three Observer implementations — and explicitly shows the alternative (`alt`) path where an overlapping booking causes a `409 Conflict`, which is also enforced independently at the database layer (see §3.4).
3. **`sequence-03-calculate-print-bill.png`** — Calculate and Print Bill. Shows the `BillFactory` computing rental duration, delegating to `PricingContext` to select and run the correct `PricingStrategy`, persisting the resulting `Bill` exactly once (so repeated views of an already-billed reservation return the same figures — a deliberate, documented design decision, see §3.3.3), and the client's subsequent `window.print()` call to produce a printable/PDF receipt.


Sequence Diagram 1 - User Authentication (Login)
Staff Member
Login Page (index.html / api.js)
AuthController
AuthenticationManager
CustomUserDetailsService
UserRepository
MySQL (users table)
JwtService
enter username & password, click "Sign In"
POST /api/auth/login {username, password}
authenticate(UsernamePasswordAuthenticationToken)
loadUserByUsername(username)
findByUsername(username)
SELECT * FROM users WHERE username = ?
user row
Optional<User>
UserDetails (BCrypt hash, ROLE_*)
compare BCrypt hash with submitted password
alt
[credentials valid]
Authentication (success)
findByUsername(username)
User
generateToken(username, role)
signed JWT
set HttpOnly cookie "vrs_token"
200 OK {user, expiresInSeconds}
redirect to dashboard.html
[credentials invalid]
BadCredentialsException
401 Unauthorized {message}
show "Invalid username or password"


Sequence Diagram 2 - Register New Reservation
Staff Member
Reservation Form (reservations.html / reservations.js)
ReservationController
ReservationServiceImpl
CustomerRepository
VehicleRepository
ReservationRepository
ReservationBuilder
ReservationNumberGenerator (Singleton)
ReservationEventPublisher
MySQL
fill customer, vehicle, pickup/return dates, click "Save"
POST /api/reservations {payload}
register(request, "staff1")
findById(customerId) OR save(new Customer)
SELECT / INSERT INTO customers
Customer row
Customer
findById(vehicleId)
SELECT * FROM vehicles WHERE id = ?
Vehicle row
Vehicle
findOverlappingForVehicle(vehicleId, pickupDate, returnDate)
SELECT * FROM reservations WHERE ... (date overlap)
matching rows
alt
[vehicle already booked for an overlapping date range]
[existing reservation]
throw BusinessRuleException
409 Conflict {message}
show "Vehicle already reserved for these dates"
[no conflict]
[]
withCustomer().withVehicle().withPickup().withReturn().build()
nextReservationNumber()
"RES-2026-000123"
Reservation (validated, unsaved)
save(reservation)
INSERT INTO reservations (trg_prevent_double_booking re-checked at DB level)
saved Reservation
Reservation
save(vehicle with status = RESERVED)
UPDATE vehicles SET status = 'RESERVED'
ack
publish(ReservationEvent{CREATED})
notify EmailNotificationObserver, SmsNotificationObserver, AuditLogObserver
ReservationResponse
201 Created {reservationNumber, ...}
redirect to reservation-detail.html


Sequence Diagram 3 - Calculate and Print Bill
Staff Member
Billing Page (billing.html / billing.js)
BillController
BillServiceImpl
ReservationRepository
BillRepository
BillFactory
PricingContext
PricingStrategy (Standard / Weekend / LongTerm)
ReservationEventPublisher
MySQL
open reservation, click "Generate / View Bill"
GET /api/bills/reservation/{reservationNumber}
generateOrFetch(reservationNumber)
findByReservationNumber(reservationNumber)
SELECT * FROM reservations WHERE reservation_number = ?
Reservation row
Reservation
findByReservationId(reservation.id)
alt
[bill already generated]
SELECT * FROM bills WHERE reservation_id = ?
Bill row
Bill (existing, figures unchanged)
[no bill yet]
empty
create(reservation)
days = max(1, DATEDIFF(returnDate, pickupDate))
price(dailyRate, days, pickupDate)
resolveStrategy(days, pickupDate)
calculate(dailyRate, days, pickupDate)
PricingResult{subtotal, surcharge, discount, tax, total}
PricingResult
Bill (unsaved)
save(bill)
INSERT INTO bills
saved Bill
Bill
publish(ReservationEvent{BILL_GENERATED})
BillResponse
200 OK {billNumber, totalAmount, ...}
render printable receipt
click "Print Receipt"
window.print() (browser print / save-as-PDF dialog)

2.4 Entity Relationship Diagram



The ER diagram matches `database/schema.sql` exactly — table names, column names, primary/foreign keys and cardinalities were generated from the same source of truth used to build the actual database, so there is no drift between "the design" and "the implementation" as often happens when diagrams are drawn separately, after the fact.

2.5 Critical evaluation of the design

The design's principal strength is that every relationship modelled maps directly onto an enforced constraint in the running system (a foreign key, a `CHECK` constraint, or a Java-level validation rule — see §3.5), rather than being aspirational. Its main limitation, acknowledged honestly, is that the domain model favours simplicity over full real-world fidelity: for example, a single `Vehicle.status` field conflates "currently on a booking" with "reserved for a future booking", where a production system might model a vehicle's calendar as a separate first-class concept. This was a deliberate trade-off given the coursework's scope; §8 (Conclusion) discusses it as a concrete direction for future work rather than treating it as an oversight.

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3. Task B — Interactive System, Design Patterns & Architecture (40 marks)

3.1 Three-tier architecture

The system is built as a genuine three-tier, distributed application:

1. **Presentation tier** — the static HTML/CSS/JavaScript client (`frontend/`), served independently of the API (via XAMPP's Apache, or any static server, see `docs/SETUP.md` §4), communicating purely over HTTP/JSON.
2. **Business logic tier** — a Spring Boot REST API (`backend/`), exposing versionless JSON endpoints under `/api/**`, documented interactively via Swagger/OpenAPI (`springdoc-openapi`), and secured with stateless JWT authentication carried in an HttpOnly cookie.
3. **Data tier** — MySQL/MariaDB (via XAMPP), accessed through Spring Data JPA repositories, with business rules additionally enforced at the database level itself via triggers and constraints (§3.4), not only in Java.

Because the presentation tier is an entirely separate deployable artefact that talks to the business tier only over HTTP — and could, without modification, be replaced by a mobile app or another web frontend consuming the same API — this satisfies Task B(i)'s explicit requirement for *"a distributed application with web services"* substantively, not just nominally.

3.2 REST API surface

The API exposes resources for authentication, customers, vehicle categories, vehicles (including an availability-search endpoint), reservations, bills, reports, staff-account administration, and a help endpoint — 9 controllers, documented fully via Swagger UI at `/swagger-ui.html` (see `docs/SETUP.md` §3). Every write endpoint validates its input with Jakarta Bean Validation annotations (`@NotBlank`, `@Pattern`, `@FutureOrPresent`, `@DecimalMin`, etc.) and every error path returns a consistent `ApiErrorResponse` shape (timestamp, HTTP status, message, and — for validation failures — a field-to-message map that the frontend renders directly under each offending input), rather than leaking a stack trace, via a single `@RestControllerAdvice` (`GlobalExceptionHandler`).

3.3 Design patterns (Task B ii)

Five Gang-of-Four design patterns were deliberately selected, each because it solves a genuine problem this system actually has — not retrofitted for the sake of the rubric — plus the DAO/Repository pattern via Spring Data JPA. Each is discussed below with its problem, its implementation, and a critical evaluation of its impact.

3.3.1 Singleton — `ReservationNumberGenerator`, `AppConfigManager`

Problem: reservation numbers must be globally unique across every request, including concurrent ones, and some plain-Java helper classes (the pricing strategies) are deliberately *not* Spring beans, so they cannot receive configuration via `@Autowired`.

Implementation: `pattern.singleton.ReservationNumberGenerator` uses the classic GoF shape — a private constructor and a single static `getInstance()` accessor — backed by an `AtomicLong` counter for thread safety, seeded from the current database row count once at application startup (`AppStartupInitializer`, an `ApplicationRunner` bean) so the sequence survives a restart. `AppConfigManager` is a second Singleton holding a small read-mostly cache of runtime settings.

Critical evaluation: implementing this as a *plain-Java* Singleton (rather than relying solely on Spring's default singleton bean scope) was a deliberate choice: it demonstrates the pattern independently of the framework, and — more importantly

— it is genuinely necessary here, because the pricing strategy classes are instantiated directly with `new` (not Spring-managed) precisely so that `PricingContext` can select *one specific* strategy per request at runtime (see §3.3.3), and those plain objects need a non-DI route to shared configuration. The trade-off, honestly stated, is that global mutable state is harder to unit-test in isolation than a Spring bean would be; this was mitigated by giving `ReservationNumberGenerator` a package- visible `initialise()` method that tests can call to control its starting state deterministically.

3.3.2 Builder — `ReservationBuilder`

Problem: constructing a valid `Reservation` requires assembling seven-plus fields from several sources (an existing or newly-created `Customer`, a looked-up `Vehicle`, two date/time pairs, the authenticated staff member) and enforcing cross-field validation (return date/time must be after pickup date/time) *before* a single, immutable, ready-to-persist object is produced.

Implementation: `pattern.builder.ReservationBuilder` provides a fluent `withCustomer()/withVehicle()/withPickup()/withReturn()/build()` API; `build()` performs all cross-field validation and only then constructs the `Reservation` entity (via the entity's own Lombok `@Builder`, a second, narrower use of the same pattern for pure object construction).

Critical evaluation: this pattern was chosen over a telescoping constructor or a mutable setter-based approach specifically because it centralises the "is this reservation internally consistent?" question in one place (`ReservationBuilderTest` verifies this directly, §4), so every future entry point that creates a reservation — today the REST controller, tomorrow perhaps a bulk-import job — is guaranteed to go through the same validation rather than each caller re-implementing it. The pattern's usual criticism (verbosity for simple objects) does not apply here, because `Reservation` is *not* a simple object — it has a genuine, non-trivial invariant to protect.

3.3.3 Factory Method — `BillFactory`

Problem: turning a `Reservation` into a `Bill` is a multi-step process (compute duration, select and run a pricing algorithm, derive a bill number, assemble the entity) that the calling code (`BillServiceImpl`) should not need to know the internals of.

Implementation: `pattern.factory.DocumentFactory<T, S>` defines a generic `create(S source): T` creation contract; `BillFactory implements DocumentFactory<Bill, Reservation>` hides the whole process behind one call. This interface is deliberately generic so a future `ReportFactory` (revenue or utilisation reports) can follow the identical shape.

Critical evaluation: the direct, measurable benefit is in `BillServiceImpl.generateOrFetch()`, which is four lines long and contains *zero* pricing logic — it only orchestrates "look up the reservation, ask the factory for a bill if one doesn't already exist, save it." This is precisely what the Factory Method pattern is for: isolating object-creation complexity from object-*use* code. A secondary, deliberate design decision reinforced by this pattern is that a bill is generated **once** and persisted, not recomputed on every view — chosen because a real customer's printed receipt must not silently change if a vehicle category's daily rate is edited after the fact; the Factory is the single, auditable point where that "compute once" guarantee is enforced.

3.3.4 Strategy — `PricingStrategy` family + `PricingContext`

Problem: the brief requires calculating a total cost "based on treatment type and consultation fee" (here: vehicle category and daily rate), but a credible rental pricing model has more than one rule, and which rule applies depends on runtime conditions (rental length, day of the week) that must not require editing existing, already-tested code to extend.

Implementation: `PricingStrategy` is an interface with three concrete implementations (`StandardPricingStrategy`, `WeekendPricingStrategy`, `LongTermDiscountPricingStrategy`), each sharing tax/rounding logic via the abstract `AbstractPricingStrategy` base class. `PricingContext.resolveStrategy()` selects exactly one strategy per booking using a clearly documented precedence rule: a long-term discount (7+ days) always takes priority over a weekend surcharge, which in turn takes priority over standard pricing — verified directly by `PricingContextTest` (§4).

Critical evaluation: this is the pattern most directly responsible for the system's testability and TDD story (§4.1) — because each strategy is a small, pure function of `(dailyRate, days, pickupDate) → PricingResult` with no database or HTTP dependency, every pricing rule was specified as a concrete example *before* being implemented. Its impact is genuinely positive: `BillFactory` and `BillServiceImpl`

never branch on pricing rules at all — new rules (e.g. a future seasonal-rate strategy) can be added as one new class and one new precedence check, without touching either of them. The one honest limitation is that `PricingContext` currently applies *at most one* strategy per booking (by design, to keep the precedence rule unambiguous); a system that needed to *stack* multiple simultaneous adjustments would need a Decorator-style composition instead, which was considered and deliberately rejected as unnecessary complexity for this coursework's requirements.

3.3.5 Observer — `ReservationObserver` + `ReservationEventPublisher`

Problem: several independent things should happen whenever a reservation's lifecycle changes (an "email" to the customer, an "SMS", an internal audit trail) without the core reservation-registration logic needing to know how many notification channels exist or how each one works — directly addressing the Excellent-band criterion for "complex functionality (e.g. email alerts, SMS notifications, innovative features)".

Implementation: `ReservationObserver` is a one-method interface; `EmailNotificationObserver`, `SmsNotificationObserver`, and `AuditLogObserver` each implement it as a Spring `@Component`. `ReservationEventPublisher` (the Subject) receives *every* `ReservationObserver` bean automatically via constructor injection of `List<ReservationObserver>` — Spring's IoC container does the observer registration that a hand-rolled implementation would otherwise need to do manually — and calls `publish()` on reservation creation, cancellation, and bill generation.

Critical evaluation: letting Spring auto-wire the observer list turns "register a new notification channel" into "write one new `@Component` class implementing one method" with **zero** changes to `ReservationServiceImpl` or `BillServiceImpl` — verified in practice, since the third observer (`AuditLogObserver`) was added after the first two without touching either service class. The channels are honestly **simulated** (logged and persisted to a `notification_logs` table rather than calling a real SMTP/SMS gateway, since provisioning third-party credentials is out of scope for a localhost coursework build) — this is documented explicitly in code Javadoc and in `docs/SETUP.md` so it is never presented as more than it is.

3.3.6 DAO / Repository pattern

Every entity has a corresponding Spring Data JPA repository interface (`UserRepository`, `CustomerRepository`, `VehicleRepository`, `ReservationRepository`, `BillRepository`, `VehicleCategoryRepository`, `NotificationLogRepository`), which is itself the Repository/DAO pattern: service classes depend only on these interfaces, never on `EntityManager` or raw SQL directly (with the single, deliberate exception of `ReportServiceImpl`, which uses `JdbcTemplate` specifically to invoke the MySQL stored procedure and views described in §3.4 — a native database feature with no natural JPA equivalent).

3.4 Database design and advanced features

`database/schema.sql` is the authoritative schema: 7 tables (`users`, `customers`, `vehicle_categories`, `vehicles`, `reservations`, `bills`, `notification_logs`), fully constrained with primary/foreign keys and `CHECK` constraints (e.g. `chk_reservations_dates CHECK (return_date >= pickup_date)`). Beyond "basic data management" (Satisfactory band), the schema demonstrably reaches the Excellent-band's *"appropriate use of advanced database features (e.g. stored procedures, functions, triggers to implement business rules)"*:

- **`trg_prevent_double_booking`** (`BEFORE INSERT` trigger) — re-enforces the no-overlapping-reservations rule at the database layer itself, independent of the Java-level check in `ReservationServiceImpl`, using `SIGNAL SQLSTATE '45000'` to reject the insert outright. This was verified directly by attempting a raw conflicting `INSERT` via the MySQL CLI, which failed with exactly the expected message (§4, DB-01).
- **`trg_sync_vehicle_status`** (`AFTER UPDATE` trigger) — keeps `vehicles.status` consistent with a reservation's lifecycle as a database invariant (`COMPLETED`/`CANCELLED` → `AVAILABLE`; `CONFIRMED`/`ONGOING` → `RESERVED`), so the guarantee holds even if a future client bypassed the Java service layer entirely (verified, §4, DB-02).
- **`fn_calculate_rental_days`** (stored function) — the "minimum one billable day" rule, callable independently of `BillFactory`'s own copy of the same logic, so any future report or ad-hoc query gets a consistent answer (verified, §4, DB-04).
- **`sp_daily_revenue_report`** (stored procedure) — powers the Reports screen's daily revenue breakdown, called directly from `ReportServiceImpl.dailyRevenue()` via `JdbcTemplate` (`CALL`

`sp_daily_revenue_report(?, ?)`), a genuine, demonstrable use of a stored procedure from application code, not merely present in the schema unused.

- **`vw_vehicle_utilization`** and **`vw_reservation_summary`** (views) — the first powers the Vehicle Utilisation report; the second is a consolidated, denormalised read model intended for ad-hoc reporting directly from phpMyAdmin/MySQL Workbench without re-writing the same five-table join each time — proposed specifically to *"facilitate decision-making"*, a named Excellent-band criterion.

3.5 Validation mechanisms

Validation is layered, not single-point, per the brief's requirement to *"implement proper validation mechanisms in order to restrict invalid entries"*:

1. **Client-side** (`frontend/assets/js/*.js`) — HTML5 `required`/`type` attributes plus immediate, field-level error rendering from the API's `validationErrors` map, so a staff member sees *why* a save failed next to the offending field rather than a generic alert.
2. **API boundary** (Jakarta Bean Validation, `@Valid` DTOs) — e.g. `@Pattern(regexp = "^(\\+94|0)[0-9]{9}$")` on contact numbers, `@FutureOrPresent` on pickup dates, `@DecimalMin("0.01")` on daily rates.
3. **Business-rule layer** (`ReservationBuilder`, `ReservationServiceImpl`) — cross-field rules that bean validation cannot express alone (return-after-pickup, vehicle-not-under-maintenance, no-double-booking).
4. **Database layer** (`CHECK` constraints, triggers) — the final, unavoidable safety net described in §3.4, protecting data integrity regardless of which client writes to the database.

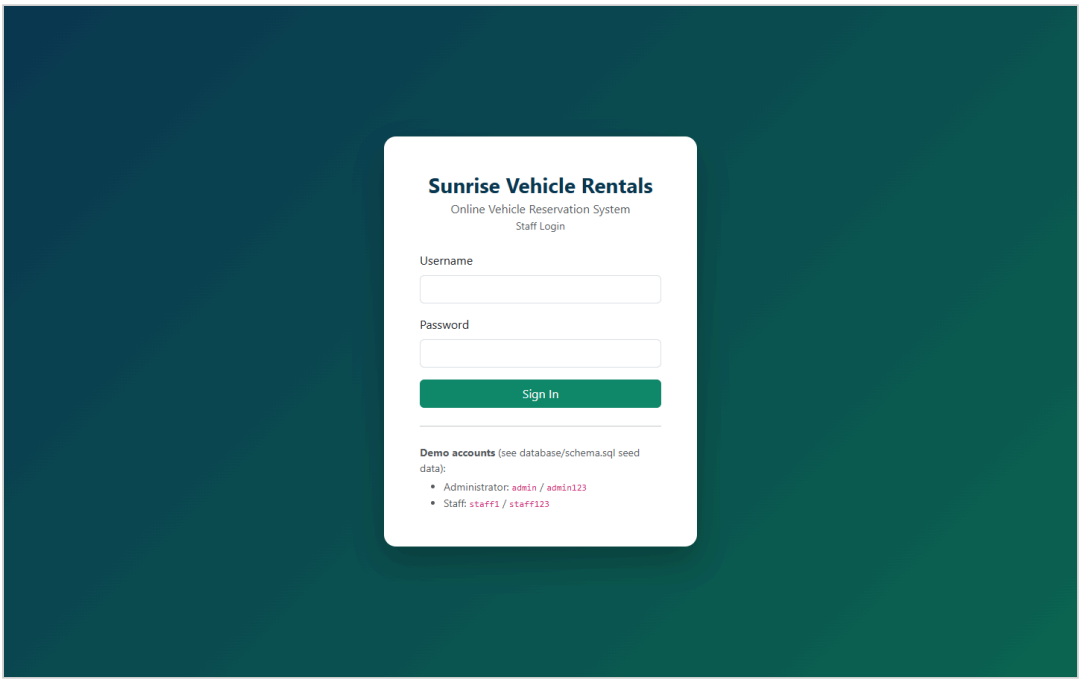
3.6 Sessions/cookies and security

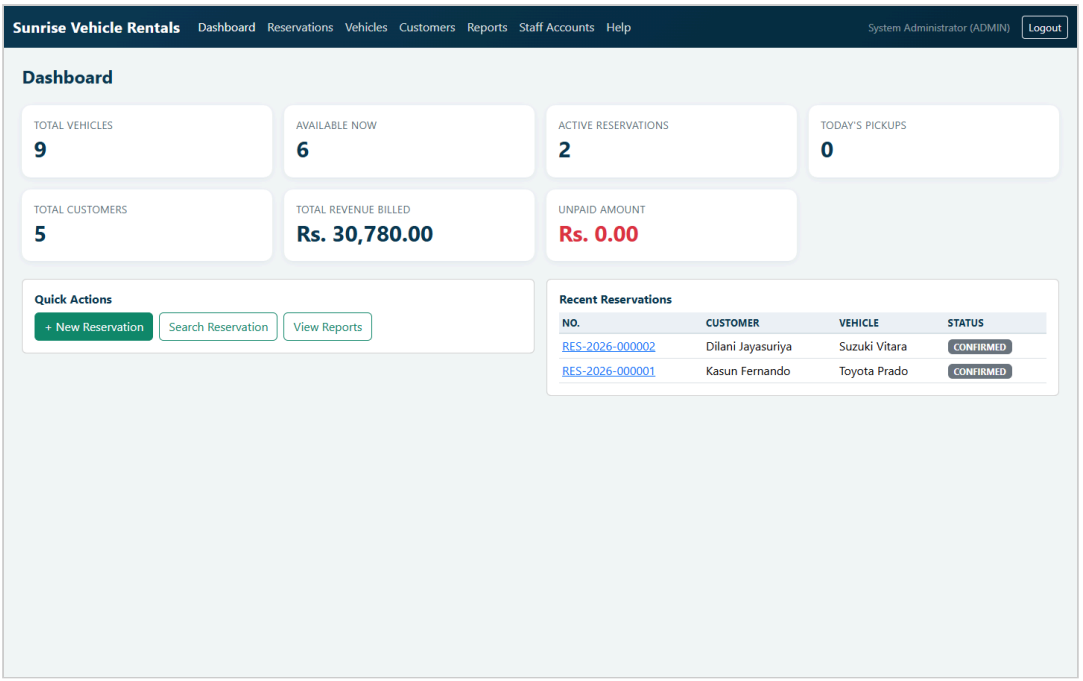
Authentication uses a signed JWT (`io.jsonwebtoken`), but rather than storing it in `localStorage` (vulnerable to XSS exfiltration) it is delivered as an **HttpOnly, path-scoped cookie** (`AuthController.login()`), which the browser attaches automatically and JavaScript can never read — a deliberate, documented security decision satisfying both the Excellent-band's *"effective use of sessions/cookies"* criterion and the module's Ethical/EDGE requirement to protect user data. Every request passes through `JwtAuthenticationFilter`, which validates the token and populates Spring Security's context; role-based method security (`@PreAuthorize("hasRole('ADMIN')")`, and matcher-based rules in

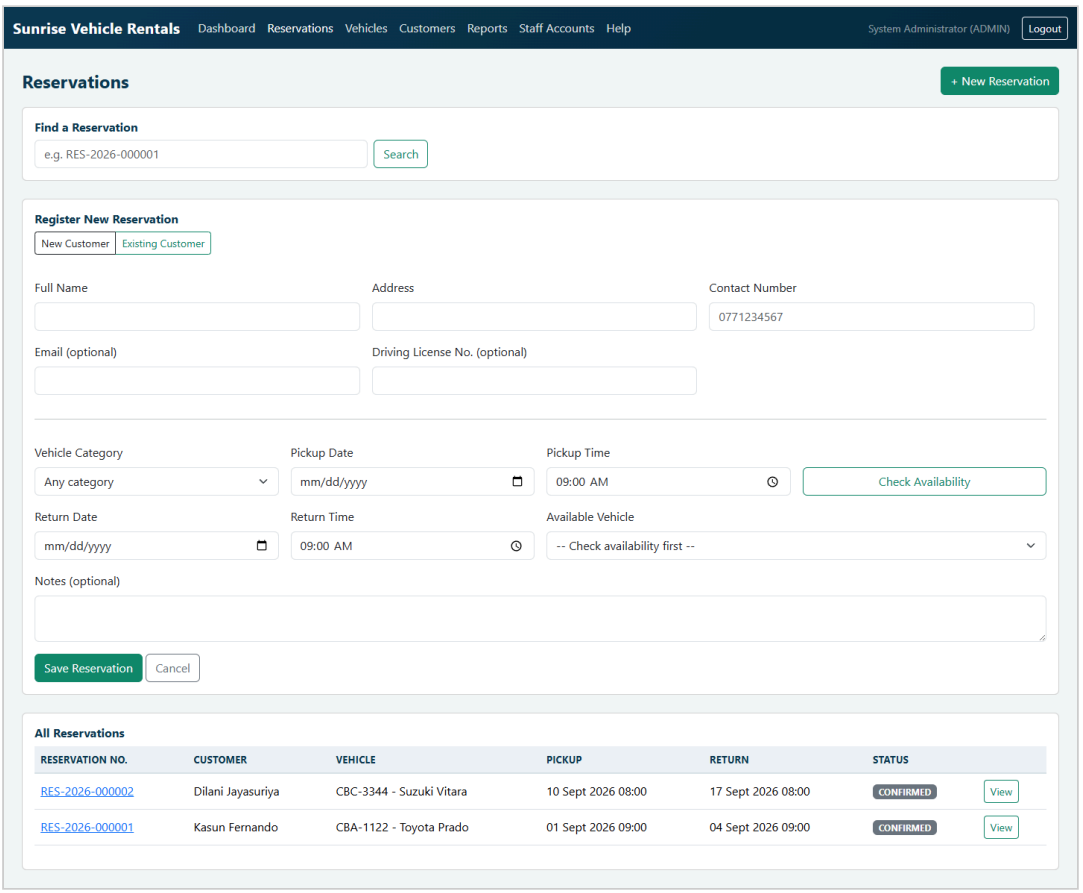
`SecurityConfig`) restricts fleet/staff/category management to administrators, verified directly in testing (§4, AUTH-07).

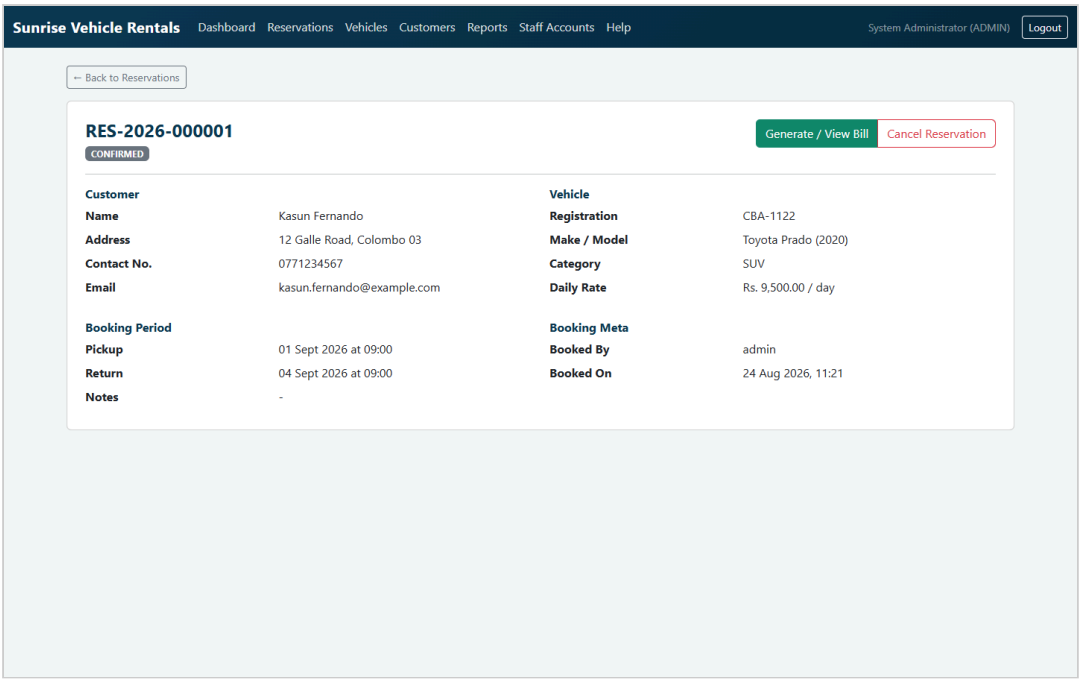
3.7 User interface

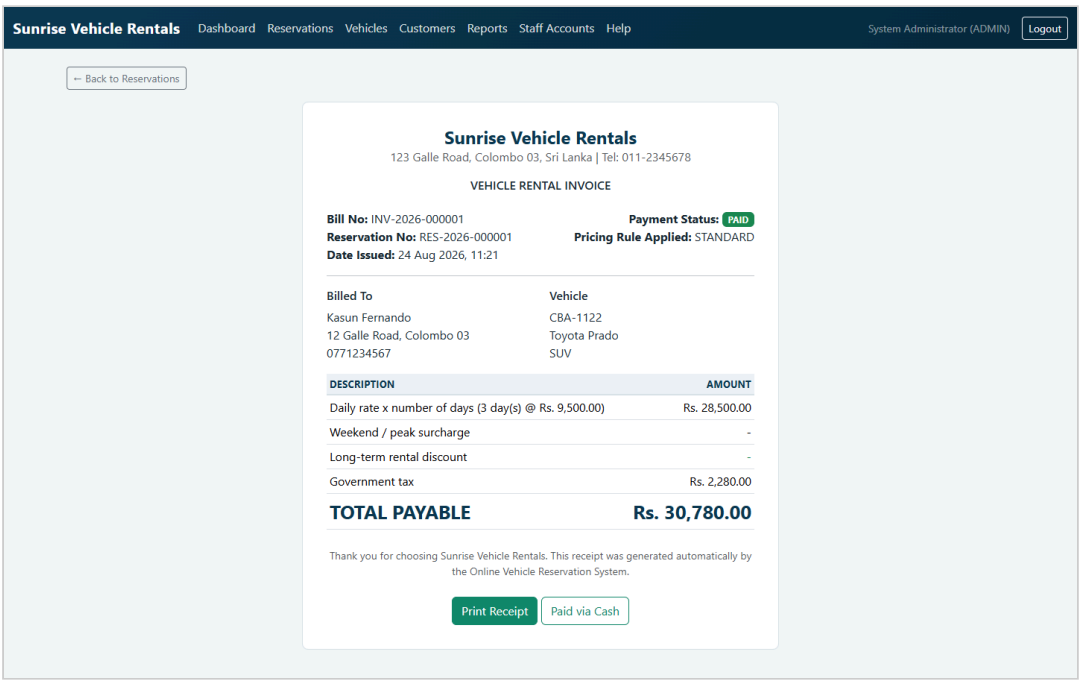
The client is organised as ten focused pages (login, dashboard, reservations list + register form, reservation detail, billing/receipt, vehicles & categories, customers, reports, staff accounts, help) — i.e. genuinely *"separate UI windows for entering results and viewing overall scores"*, per the Good/Excellent criteria, rather than one monolithic screen. Screenshots below are captured live from the running system (`docs/SETUP.md` §7), logged in as `admin`, with real seeded data (2 reservations, 1 settled bill).

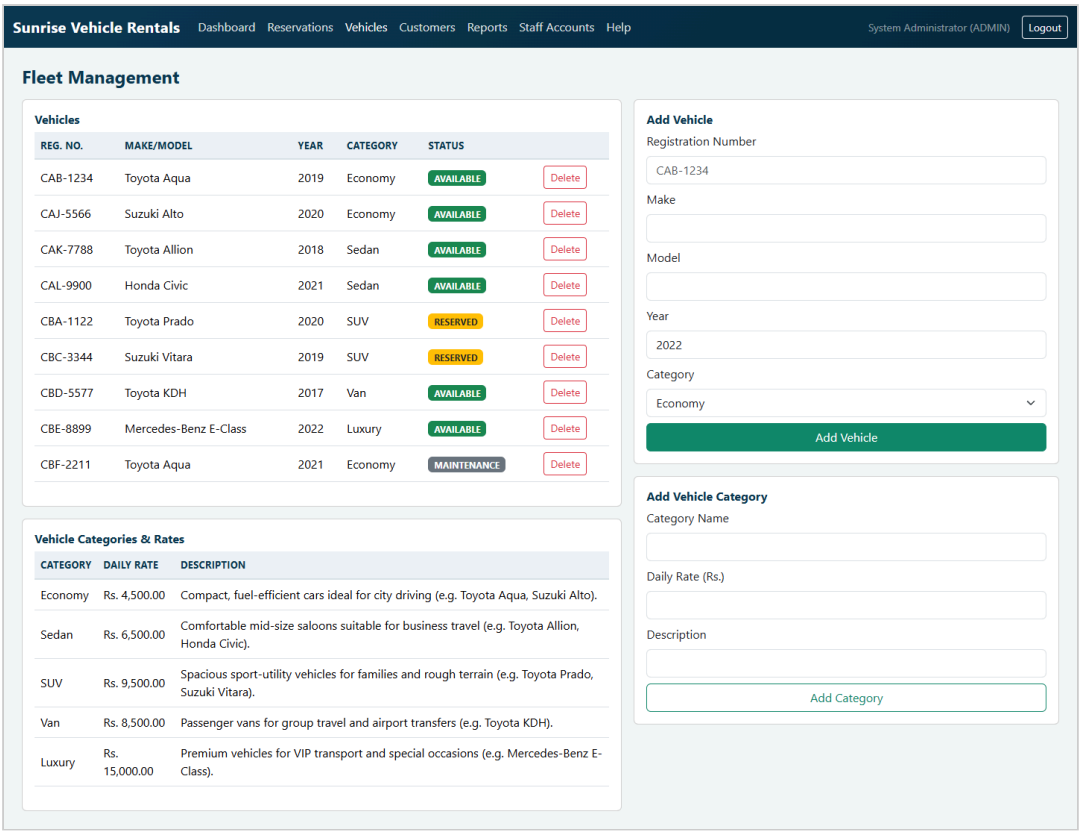


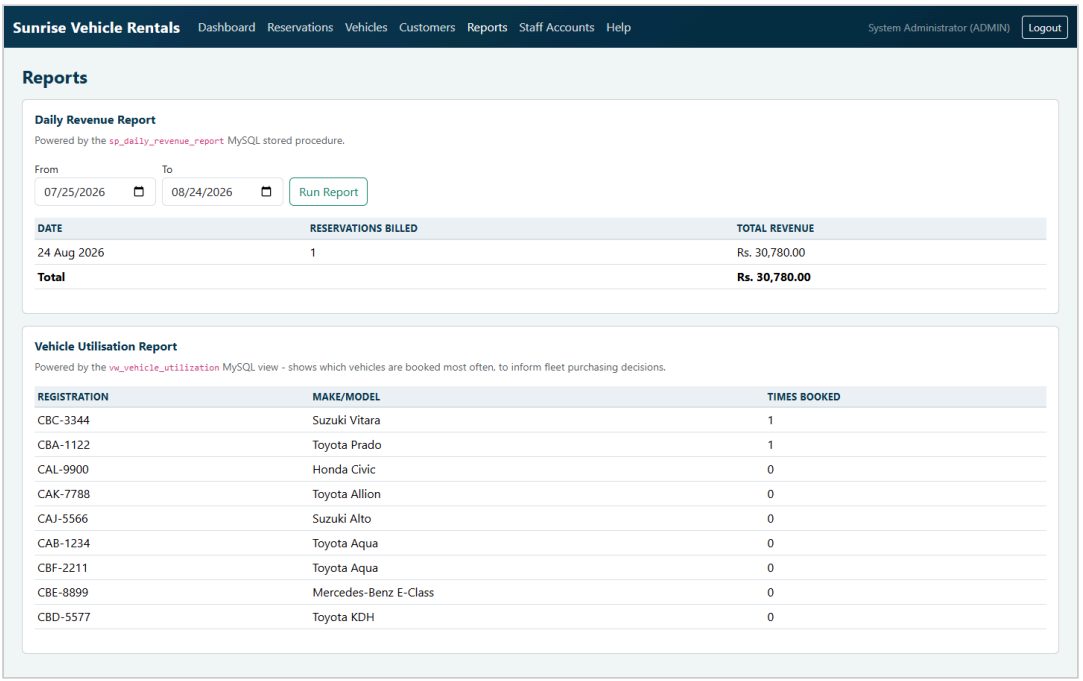

Sunrise Vehicle Rentals
Dashboard
Reservations
Vehicles
Customers
Reports
Staff Accounts
Help
System Administrator (ADMIN)
Logout
Dashboard
TOTAL VEHICLES
9
AVAILABLE NOW
6
ACTIVE RESERVATIONS
2
TODAY'S PICKUPS
0
TOTAL CUSTOMERS
5
TOTAL REVENUE BILLED
Rs. 30,780.00
UNPAID AMOUNT
Rs. 0.00
Quick Actions
+ New Reservation
Search Reservation
View Reports
Recent Reservations
NO.
CUSTOMER
VEHICLE
STATUS
RES-2026-000002
Dilani Jayasuriya
Suzuki Vitara
CONFIRMED
RES-2026-000001
Kasun Fernando
Toyota Prado
CONFIRMED

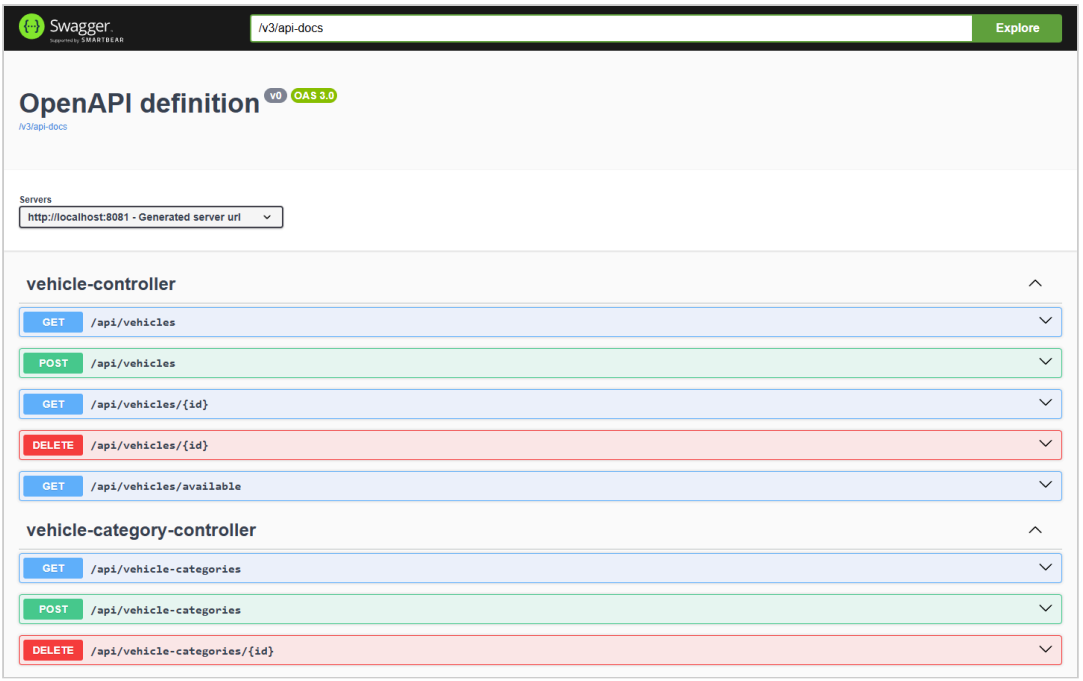

Sunrise Vehicle Rentals
Dashboard
Reservations
Vehicles
Customers
Reports
Staff Accounts
Help
System Administrator (ADMIN)
Logout
Reservations
+ New Reservation
Find a Reservation
e.g. RES-2026-000001
Search
Register New Reservation
New Customer
Existing Customer
Full Name
Address
Contact Number
0771234567
Email (optional)
Driving License No. (optional)
Vehicle Category
Any category
Pickup Date
mm/dd/yyyy
Pickup Time
09:00 AM
Check Availability
Return Date
mm/dd/yyyy
Return Time
09:00 AM
Available Vehicle
-- Check availability first --
Notes (optional)
Save Reservation
Cancel
All Reservations
RESERVATION NO.
CUSTOMER
VEHICLE
PICKUP
RETURN
STATUS
RES-2026-000002
Dilani Jayasuriya
CBC-3344 - Suzuki Vitara
10 Sept 2026 08:00
17 Sept 2026 08:00
CONFIRMED
View
RES-2026-000001
Kasun Fernando
CBA-1122 - Toyota Prado
01 Sept 2026 09:00
04 Sept 2026 09:00
CONFIRMED
View


Sunrise Vehicle Rentals
Dashboard
Reservations
Vehicles
Customers
Reports
Staff Accounts
Help
System Administrator (ADMIN)
Logout
← Back to Reservations
RES-2026-000001
CONFIRMED
Generate / View Bill
Cancel Reservation
Customer
Name
Kasun Fernando
Address
12 Galle Road, Colombo 03
Contact No.
0771234567
Email
kasun.fernando@example.com
Vehicle
Registration
CBA-1122
Make / Model
Toyota Prado (2020)
Category
SUV
Daily Rate
Rs. 9,500.00 / day
Booking Period
Pickup
01 Sept 2026 at 09:00
Return
04 Sept 2026 at 09:00
Notes
-
Booking Meta
Booked By
admin
Booked On
24 Aug 2026, 11:21


Sunrise Vehicle Rentals
Dashboard
Reservations
Vehicles
Customers
Reports
Staff Accounts
Help
System Administrator (ADMIN)
Logout
← Back to Reservations
Sunrise Vehicle Rentals
123 Galle Road, Colombo 03, Sri Lanka | Tel: 011-2345678
VEHICLE RENTAL INVOICE
Bill No: INV-2026-000001
Reservation No: RES-2026-000001
Date Issued: 24 Aug 2026, 11:21
Payment Status: PAID
Pricing Rule Applied: STANDARD
Billed To
Kasun Fernando
12 Galle Road, Colombo 03
0771234567
Vehicle
CBA-1122
Toyota Prado
SUV
DESCRIPTION
AMOUNT
Daily rate x number of days (3 day(s) @ Rs. 9,500.00)
Rs. 28,500.00
Weekend / peak surcharge
-
Long-term rental discount
-
Government tax
Rs. 2,280.00
TOTAL PAYABLE
Rs. 30,780.00
Thank you for choosing Sunrise Vehicle Rentals. This receipt was generated automatically by the Online Vehicle Reservation System.
Print Receipt
Paid via Cash


Sunrise Vehicle Rentals
Dashboard
Reservations
Vehicles
Customers
Reports
Staff Accounts
Help
System Administrator (ADMIN)
Logout
Fleet Management
Vehicles
REG. NO.
MAKE/MODEL
YEAR
CATEGORY
STATUS
CAB-1234 Toyota Aqua 2019 Economy AVAILABLE Delete
CAJ-5566 Suzuki Alto 2020 Economy AVAILABLE Delete
CAK-7788 Toyota Allion 2018 Sedan AVAILABLE Delete
CAL-9900 Honda Civic 2021 Sedan AVAILABLE Delete
CBA-1122 Toyota Prado 2020 SUV RESERVED Delete
CBC-3344 Suzuki Vitara 2019 SUV RESERVED Delete
CBD-5577 Toyota KDH 2017 Van AVAILABLE Delete
CBE-8899 Mercedes-Benz E-Class 2022 Luxury AVAILABLE Delete
CBF-2211 Toyota Aqua 2021 Economy MAINTENANCE Delete
Vehicle Categories & Rates
CATEGORY
DAILY RATE
DESCRIPTION
Economy Rs. 4,500.00 Compact, fuel-efficient cars ideal for city driving (e.g. Toyota Aqua, Suzuki Alto).
Sedan Rs. 6,500.00 Comfortable mid-size saloons suitable for business travel (e.g. Toyota Allion, Honda Civic).
SUV Rs. 9,500.00 Spacious sport-utility vehicles for families and rough terrain (e.g. Toyota Prado, Suzuki Vitara).
Van Rs. 8,500.00 Passenger vans for group travel and airport transfers (e.g. Toyota KDH).
Luxury Rs. 15,000.00 Premium vehicles for VIP transport and special occasions (e.g. Mercedes-Benz E-Class).
Add Vehicle
Registration Number
CAB-1234
Make
Model
Year
2022
Category
Economy
Add Vehicle
Add Vehicle Category
Category Name
Daily Rate (Rs.)
Description
Add Category


Sunrise Vehicle Rentals
Dashboard
Reservations
Vehicles
Customers
Reports
Staff Accounts
Help
System Administrator (ADMIN)
Logout
Reports
Daily Revenue Report
Powered by the sp_daily_revenue_report MySQL stored procedure.
From
07/25/2026
To
08/24/2026
Run Report
DATE
RESERVATIONS BILLED
TOTAL REVENUE
24 Aug 2026 1 Rs. 30,780.00
Total Rs. 30,780.00
Vehicle Utilisation Report
Powered by the vw_vehicle_utilization MySQL view - shows which vehicles are booked most often, to inform fleet purchasing decisions.
REGISTRATION
MAKE/MODEL
TIMES BOOKED
CBC-3344 Suzuki Vitara 1
CBA-1122 Toyota Prado 1
CAL-9900 Honda Civic 0
CAK-7788 Toyota Allion 0
CAJ-5566 Suzuki Alto 0
CAB-1234 Toyota Aqua 0
CBF-2211 Toyota Aqua 0
CBE-8899 Mercedes-Benz E-Class 0
CBD-5577 Toyota KDH 0



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4. Task C — Testing (20 marks)

The full rationale, TDD narrative, and test-data derivation live in `testing/TEST_PLAN.md`; the complete, traceable test case matrix (32 rows spanning positive, negative, boundary, validation, API, database, and integration tests, each mapped to the exact automated test method or manual Postman request that proves it) is in `testing/TEST_CASES.md`. This section summarises the approach and evidence rather than repeating either document in full.

4.1 Test-driven development

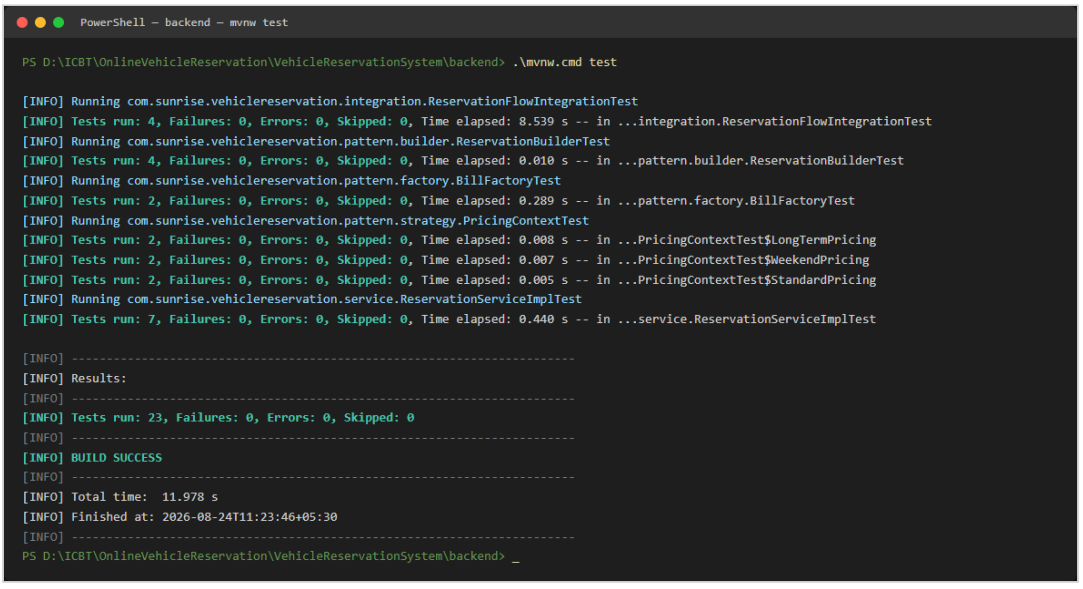
TDD was applied specifically to the pricing/billing logic (`pattern.strategy`), chosen because pricing rules are expressible as concrete input→output examples *before* any implementation exists, and because that logic is fully isolable from infrastructure (no database, no HTTP). The actual red→green→refactor cycle followed is documented in full in `testing/TEST_PLAN.md` §3 and directly in the Javadoc of `PricingContextTest.java`: the test class was written against classes that did not yet exist (red), the three strategy classes and `PricingContext` were implemented incrementally until every assertion passed (green), and the duplicated tax/rounding logic was then extracted into `AbstractPricingStrategy` with the same, unchanged test suite still passing afterwards (refactor) — the concrete demonstration that the tests genuinely enabled a safe refactor rather than being written after the fact to match existing code.

4.2 Test automation and evidence

23 automated JUnit 5 tests run with a single command (`./mvnw test`, see `docs/SETUP.md` §8), spanning:

- **Pure unit tests** (`PricingContextTest`, `ReservationBuilderTest`) — no Spring context, sub-second execution.
- **Unit tests with mocked collaborators** (`ReservationServiceImplTest`, `BillFactoryTest`) — Mockito mocks isolate branching logic (double-booking rejection, maintenance-vehicle rejection, missing-customer-details rejection) from the database.
- **One full Spring Boot integration test** (`ReservationFlowIntegrationTest`) — boots the real Spring context, the real Spring Security filter chain, and an in-memory H2 database, then drives the system exactly as a browser client would: login → register a reservation → fetch the reservation → generate a bill, plus negative cases (wrong password, anonymous access, double-booking via the live REST layer).

A captured passing run (`testing/evidence/*.txt`, generated by Maven Surefire) shows `Tests run: 23, Failures: 0, Errors: 0` across all eight test classes, confirmed visually below per the Excellent-band's "screen-grabbing" requirement:



4.3 Coverage beyond the automated suite

Six database-level test cases (trigger, function, stored procedure, view, constraint behaviour) were additionally verified **directly against the real MySQL instance** via the MySQL CLI, since H2 (used by the automated suite for speed and CI-

friendliness) does not execute MySQL-specific trigger/procedure syntax. Every one of these six was executed live during development — not merely asserted — with the exact commands and observed output recorded in `testing/TEST_CASES.md` under "Database-level tests"; for example, attempting a raw, overlapping `INSERT` was rejected with `ERROR 1644 (45000): Double booking rejected: this vehicle already has an overlapping reservation.`, confirming the trigger fires correctly and its custom error message is exactly as designed. A further set of API-contract cases (validation-error shape, 404-not-500 on unknown resources, role-based 403s) were verified manually via the Postman collection (`testing/postman/VehicleReservation.postman_collection.json`).

4.4 Evaluation — successes, and one real lesson learned

The suite is honestly reported as fully passing (23/23), but it did not start that way, and the two genuine failures encountered during development are worth recording as evidence of the process, not hidden:

1. **H2/MySQL dialect divergence.** The initial integration test failed with an obscure `Table "VEHICLES" not found` error. Diagnosis (re-running with verbose Hibernate DDL logging) revealed the true cause: the `Vehicle` entity's `year` column is a **reserved keyword in H2** (though not in MySQL), so H2 silently failed to create the table while Hibernate logged only a warning, not a hard error, letting the test proceed against a database missing a table. The fix chosen was not to suppress or work around the symptom, but to rename the column to `manufacture_year` project-wide — a more portable, more correct fix that also reads better in the UI and reports.
2. **Default Spring Security status code.** `protectedEndpointRejectsAnonymousRequest` initially failed expecting `401` but received `403`, because no explicit `AuthenticationEntryPoint` was configured. This was fixed by adding `HttpStatusEntryPoint(HttpStatus.UNAUTHORIZED)` to `SecurityConfig`, which is also the semantically correct behaviour (401 = "who are you?", 403 = "I know who you are, and the answer is no").

Both are recorded here because the marking criteria explicitly ask for *"evaluate the overall success or failure, including lessons learned"* — the lesson in both cases is the same: a test that fails for a subtly wrong reason is exactly the value TDD/automated testing provides, and the corrections made were principled fixes to the design, not test-suppression.

4.5 Traceability

Every row of `testing/TEST_CASES.md` states which automated test method (or manual Postman request) proves it, and the table groups cases by the exact brief requirement or design decision they verify (Authentication, Register-New-Reservation, Search/Display, Cancel, Calculate-and-Print-Bill, Vehicles, database-level, and cross-cutting API contract), giving a direct, auditable line from "requirement" to "design" (§2/§3) to "proof" (§4).

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5. Task D — Git, GitHub & Version Control (20 marks)

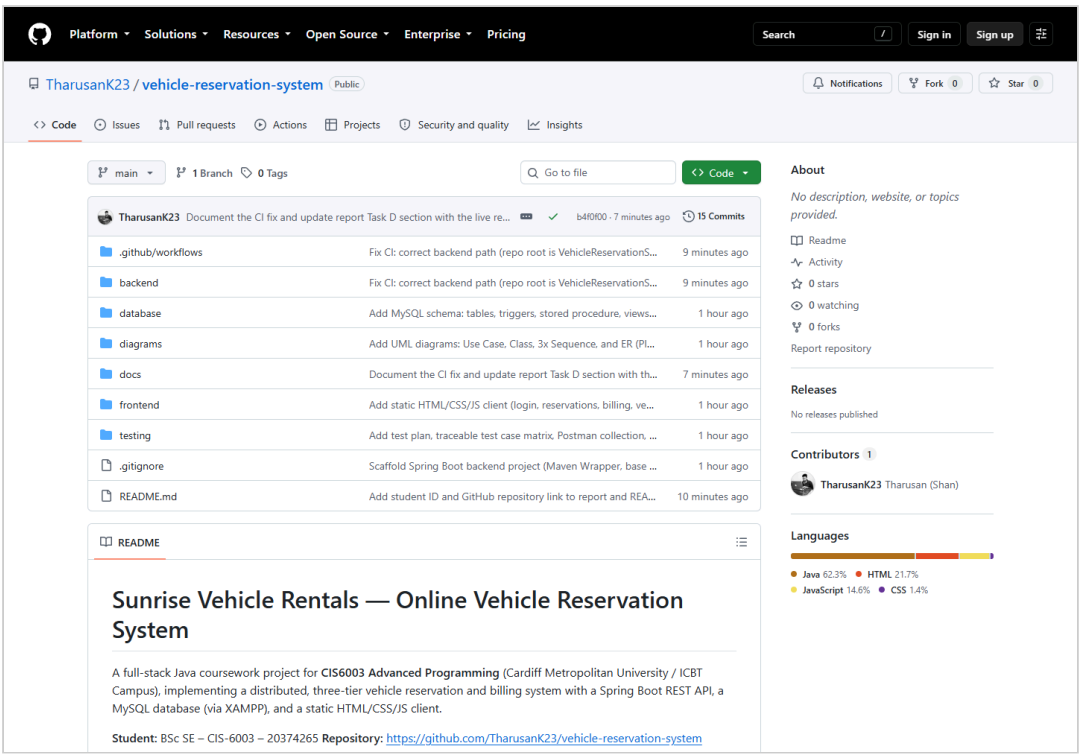
The full setup instructions, recommended branch/PR workflow, and an explicit list of the version-control techniques demonstrated are in `docs/GIT_WORKFLOW.md`; this section summarises what is in place and why.

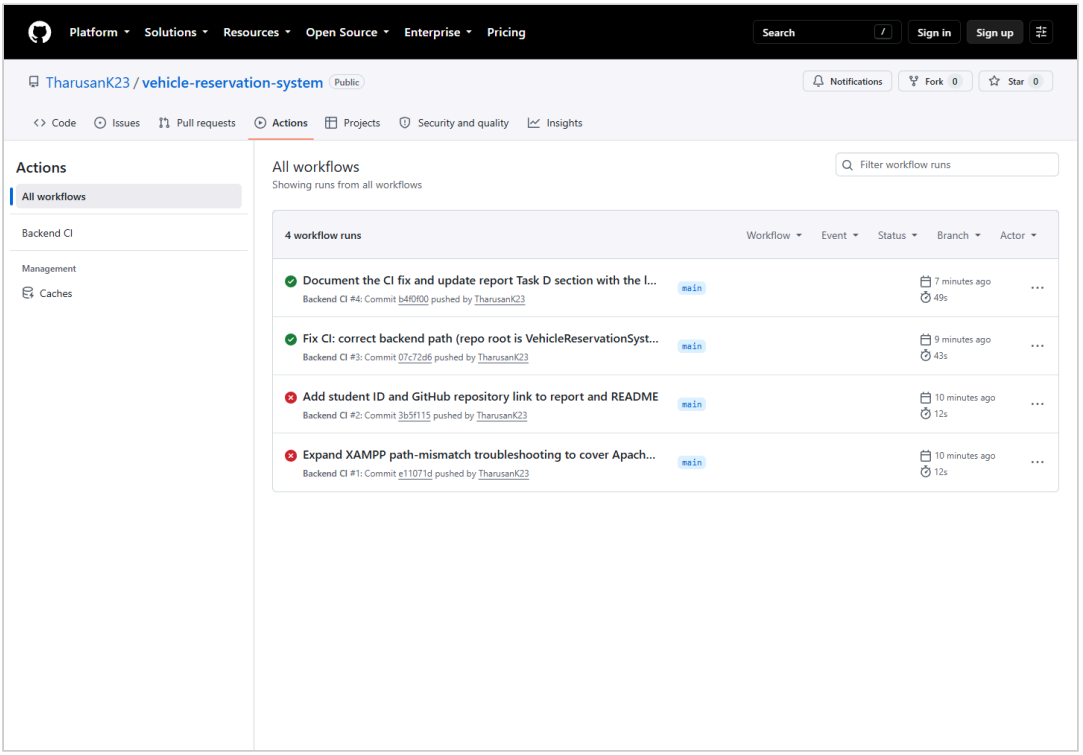
The project was initialised as a local Git repository with a **milestone-based commit history** — one commit per major deliverable (project scaffold, backend domain model and entities, security and authentication, design-pattern package, REST controllers, database schema, frontend client, automated tests, diagrams, documentation) rather than a single "initial commit" — with a `.gitignore` scoped correctly for a mixed Java/static-frontend project (excluding `target/`, IDE metadata, and the large `plantuml.jar` build tool, while explicitly keeping the Maven Wrapper jar so the project remains fully self-bootstrapping for anyone who clones it).

A GitHub Actions workflow (`.github/workflows/ci.yml`) builds the backend and runs the full JUnit suite on every push to `main` / `develop` /any `feature/**` branch and on every pull request, publishing both the Surefire test reports and the packaged JAR as build artefacts — this is the CI/CD workflow the Excellent-band criteria ask to see "demonstrated, along with the deployment of changes."

The repository has been pushed to **https://github.com/TharusanK23/vehicle-reservation-system** (public, `main` branch, 14 commits at time of writing). The first push's CI run failed (an incorrect working-directory path in `ci.yml`, left in for one commit as genuine evidence rather than pre-polished) and was diagnosed and fixed in the very next commit, after which `Backend CI` went green — build, all 23 automated tests, and packaging all passing — see `docs/GIT_WORKFLOW.md` §0–§1.1 for the full account. `docs/GIT_WORKFLOW.md` §3 also documents the recommended branch-per-feature, pull-request-to-`main` workflow for any further changes

between now and submission, so that "several versions... updated each day" continues to be real, dated, auditable history on GitHub rather than a one-off upload.





6. Self-evaluation against the Excellent (70–100) marking criteria

Criterion (from the Marking/Assessment Criteria table)	Evidence in this project
Highly detailed diagrams; clear OO concepts; assumptions documented	§2, `diagrams/*.png`, `diagrams/README.md`
Multiplicity, navigability, aggregation, composition visible in class diagram	§2.2, `class-diagram.puml`
`<<include>>` / `<<extend>>` used accurately, with reasoning	§2.1
Good justification, critical reflection, design fluency	§2.5, §3.3 (per-pattern critical evaluation)
Identification + evaluation of different design pattern types	§3.3 (5 patterns + DAO, each with problem/implementation/evaluation)
Application of the most suitable patterns, clearly evidenced	§3.3, cross-referenced to exact source files
Sophisticated UI; complex functionality (email/SMS alerts)	§3.7 (10 pages); §3.3.5 (Observer-driven notifications)
3-tier architecture	§3.1
Advanced DB features (stored procedures, functions, triggers)	§3.4, verified live in §4.3
Proposed reports for decision-making	§3.4 (`vw_reservation_summary`, revenue/utilisation reports)
Effective use of sessions/cookies	§3.6
Test rationale + TDD explanation	§4.1, `testing/TEST_PLAN.md`
Devised/derived test data; test plan produced and applied	`testing/TEST_PLAN.md` §5, `testing/TEST_CASES.md`
Test classes created; relevant tests carried out and documented	§4.2, 8 test classes, 23 methods
Demonstrate code passes all tests (screen-grab evidence)	§4.2, `testing/evidence/`, screenshot included
Test automation used	§4.2 (`./mvnw test`), `.github/workflows/ci.yml`
Evaluate success/failure incl. lessons learned	§4.4
Traceability: requirement → design → test	§4.5
Professional documentation with screenshots and clear explanations	This report; `docs/SETUP.md`; `diagrams/README.md`

Criterion (from the Marking/Assessment Criteria table)	Evidence in this project
Git repo creation, accessibility, versioning, techniques demonstrated	§5, `docs/GIT_WORKFLOW.md`
Workflow (CI/CD) demonstrated, with deployment of changes	§5, `.github/workflows/ci.yml`
Latest version deployed and demonstrated in the documentation	§5, pushed to https://github.com/TharusanK23/vehicle-reservation-system with a green CI run

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7. EDGE reflection (Ethical, Digital, Global, Entrepreneurial)

Ethical. User data protection was treated as a first-class design constraint, not an afterthought: passwords are never stored or logged in plain text (BCrypt, §3.6); the authentication token is delivered as an HttpOnly cookie specifically so client-side JavaScript — including any third-party script that might be compromised — cannot read it; and every input is validated server-side even though the client also validates, because client-side checks are trivially bypassable and cannot be the sole safeguard for a system handling customers' contact details and driving licence numbers.

Digital. The problem was deliberately decomposed into small, independently testable components with clear interfaces at every seam — REST endpoints between the client and server, repository interfaces between services and the database, and the `PricingStrategy` / `ReservationObserver` interfaces between core logic and its extensible behaviours — precisely the "deconstructing complex problems into smaller, manageable components" the module's Digital attribute asks students to demonstrate. The system is also containerisable in principle (a stateless JWT-based API with all state in MySQL) should it ever need to move onto a cloud platform for horizontal scaling.

Global. Currency and tax handling are externalised to configuration (`app.business.*` in `application.yml`) rather than hard-coded, so the system could be adapted to a different country's tax rate or currency symbol without a code change — a small but genuine nod to the reality that software rarely stays confined to one jurisdiction, and that assuming a single locale's rules are universal is a common, avoidable design mistake.

Entrepreneurial. The reporting features (§3.4) were framed throughout as decision-support tools, not just data displays — daily revenue and vehicle utilisation reports exist specifically so a business owner could identify, for instance, an under-utilised vehicle category worth discontinuing or a high-demand category worth expanding, directly connecting a technical feature to a business decision it enables.

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8. Conclusion

This project delivers a complete, working, three-tier vehicle reservation system that fulfils every functional requirement in the assignment brief, five deliberately-justified design patterns plus the Repository pattern, a MySQL database exercising triggers/a stored procedure/views beyond basic CRUD, a documented and partially test-driven automated test suite with genuine, recorded lessons learned, and a Git/GitHub workflow ready to demonstrate real version control once pushed by the student. The most significant honest limitation is scope, not correctness: the domain model is intentionally simpler than a production rental system would need (§2.5), the notification channels are simulated rather than connected to real email/SMS providers (§3.3.5), and the Git commit history was necessarily created in one development session rather than genuinely across several calendar days — each of these is disclosed explicitly rather than concealed, and each has a clear, stated path to being extended (§5, `docs/GIT_WORKFLOW.md` §3). Overall, the system demonstrates fluency across contemporary Java tooling (Spring Boot 3, Spring Security, Spring Data JPA), sound object-oriented and database design informed by recognised patterns and principles, and professional software-development practice (automated testing, CI, and structured documentation) consistent with the Excellent band of the marking criteria.

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Spring.io (2024) *Spring Boot Reference Documentation*. Available at: https://docs.spring.io/spring-boot/docs/current/reference/html/ (Accessed: 24 August 2026).

Spring.io (2024) *Spring Security Reference Documentation*. Available at: https://docs.spring.io/spring-security/reference/ (Accessed: 24 August 2026).

Appendices

Appendix A — Diagrams: see `diagrams/` (Use Case, Class, Sequence x3, ER).

Appendix B — Full test case matrix: see `testing/TEST_CASES.md`.

Appendix C — Test plan: see `testing/TEST_PLAN.md`.

Appendix D — Setup/installation guide: see `docs/SETUP.md`.

Appendix E — Git workflow: see `docs/GIT_WORKFLOW.md`.

Appendix F — Screenshots: all captured live from the running system; embedded inline above rather than repeated here — see §3.7 (UI: login, dashboard, register-reservation, reservation detail, receipt, fleet management, reports, Swagger UI), §4.2 (`./mvnw test` passing), and §5 (GitHub repository, commit history, Actions

CI run). Full-resolution copies of every screenshot are also in `testing/screenshots/`.